EMPLOYEE ATTRITION PREDICTION FOR HR USING MACHINE LEARNING

Submitted By

Ayan Majumdar

Project Type

Machine Learning Internship Project

ABSTRACT

Employee attrition is one of the major challenges faced by organizations. High attrition rates can lead to increased recruitment costs, reduced productivity, and loss of experienced employees. This project aims to predict whether an employee is likely to leave the organization using machine learning techniques.

The IBM HR Analytics Employee Attrition dataset was used for analysis and prediction. Data preprocessing, visualization, feature encoding, and machine learning algorithms were applied to build the predictive model. A Random Forest Classifier was used to predict employee attrition. The model achieved an accuracy of 86.73%, indicating its effectiveness in predicting employee turnover.

This project can help HR departments identify employees at risk of leaving and develop retention strategies accordingly.

INTRODUCTION

Employee attrition refers to the reduction of workforce due to employees leaving an organization voluntarily or involuntarily. High employee attrition can negatively affect organizational performance and increase hiring and training costs.

Predicting employee attrition enables organizations to take preventive measures to improve employee satisfaction and retention. Machine learning techniques provide an effective solution for analyzing employee data and identifying factors that contribute to attrition.

This project uses machine learning algorithms to predict whether an employee is likely to leave the company based on various employee-related factors.

PROBLEM STATEMENT

Organizations often struggle to identify employees who are likely to leave the company. Employee turnover results in financial losses, reduced productivity, and disruption of business operations.

The objective of this project is to develop a machine learning model capable of predicting employee attrition based on historical employee data.

OBJECTIVES

The objectives of this project are:

- To analyze employee data
- To identify factors influencing employee attrition
- To preprocess and clean the dataset
- To build a machine learning prediction model
- To evaluate model performance
- To assist HR departments in employee retention planning

TECHNOLOGIES USED

Technology	Purpose
Python	Programming Language
Pandas	Data Analysis
NumPy	Numerical Computation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Machine Learning
Jupyter Notebook	Development Environment

DATASET DESCRIPTION

IBM HR Analytics Employee Attrition Dataset

The dataset contains employee-related information such as:

- Age
- Gender
- Education
- Job Role
- Monthly Income
- Years at Company
- Work-Life Balance
- Job Satisfaction
- Overtime
- Attrition Status

The target variable is Attrition, which indicates whether an employee left the organization or stayed.

METHODOLOGY

The project follows the following methodology:

1. Data Collection

The IBM HR Analytics Employee Attrition dataset was collected and loaded into Python using Pandas.

2. Data Preprocessing

Data preprocessing steps included:

- Checking missing values
- Handling categorical variables
- Encoding textual data using Label Encoding
- Feature selection

3. Exploratory Data Analysis

Data visualization techniques were used to understand:

- Employee attrition distribution
- Employee demographics
- Relationships among variables

4. Model Building

A Random Forest Classifier was selected for prediction because of its high accuracy and robustness.

5. Model Evaluation

The model was evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

MACHINE LEARNING ALGORITHM

Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy.

Advantages of Random Forest

- High prediction accuracy
- Handles large datasets efficiently
- Reduces overfitting
- Suitable for classification tasks
- Handles categorical and numerical data

The Random Forest model was trained using employee data and used to predict employee attrition.

IMPLEMENTATION

The implementation process included:

1. Importing required libraries
2. Loading the dataset
3. Data preprocessing
4. Label encoding
5. Splitting data into training and testing sets
6. Training Random Forest Classifier
7. Predicting employee attrition
8. Evaluating model performance

RESULTS

The machine learning model successfully predicted employee attrition.

Accuracy Achieved

Accuracy = 86.73%

Important Factors Affecting Attrition

- Overtime
- Monthly Income
- Job Role
- Years at Company
- Job Satisfaction
- Work-Life Balance

The generated confusion matrix and feature importance graph helped evaluate the model's effectiveness.

FINAL PREDICTION

The trained model predicts whether an employee is likely to leave the company.

Example Output

Employee is likely to leave the company.

The prediction is based on employee characteristics and historical HR data.

ADVANTAGES OF THE PROJECT

- Helps HR departments identify at-risk employees
- Improves employee retention strategies
- Reduces recruitment and training costs
- Supports data-driven decision making
- Enhances workforce planning

LIMITATIONS

- Prediction quality depends on dataset quality
- External factors affecting attrition are not considered
- Model performance may vary for different organizations

FUTURE SCOPE

Future improvements may include:

- Deep Learning based prediction models
- Real-time employee monitoring systems
- Web-based HR analytics dashboards
- Cloud deployment
- Advanced employee sentiment analysis

CONCLUSION

This project demonstrates how machine learning can be used to predict employee attrition effectively. Using the IBM HR Analytics dataset and Random Forest Classifier, the model achieved an accuracy of 86.73%.

The developed system can help organizations proactively identify employees who are likely to leave and implement effective retention strategies. This contributes to improved workforce management and organizational performance.

REFERENCES

1. IBM HR Analytics Employee Attrition Dataset
2. Python Documentation
3. Scikit-learn Documentation
4. Pandas Documentation
5. Machine Learning Research Articles